

JASHANPREET SINGH

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Amritsar, Punjab, India

Education

Guru Nanak Dev University - BSc. Physics - SGPA 8.6/10 2024 – Present
Amritsar, Punjab, India

The Millennium School - Non Medical - 84.2% 2022 – 2024
Patiala, Punjab, India

AI Engineering & Technical Experience

Sitee | Lead AI Engineer & Founder 2026

- Designed and developed an AI-powered application leveraging a custom 1.2B+ parameter LLM trained on IITD Cluster and IUCAA Pegasus Cluster to autonomously generate full-stack websites, streamlining developer workflows with a brutalist-inspired UI.
- Integrated vector databases (FAISS/Pinecone) for efficient embedding storage and retrieval, ensuring scalable AI workflows and rapid prompt execution.
- www.sitee.in

PRIN | Tor Project Summer Intern 2025

- Developed a scalable intelligence platform to crawl dark web onion hosts, applying advanced NLP algorithms and prompt engineering to extract and classify leaked credentials (BTC IDs, PGP keys, IPs).
- Engineered backend automation to securely clear cache and database records upon server initialization, maintaining data integrity for real-time cyber threat ingestion.
- Optimized AI model performance for rapid threat categorization, utilizing REST APIs to feed structured intelligence data into a centralized dashboard for law enforcement.
- www.github.com/dingra11/beest_prin

Texwolf | Software Developer 2025

- Developed a hybrid free software platform integrating backend parsing systems to streamline the formatting and authoring of academic research papers natively in LaTeX.
- www.texwolf.com

Alzheimer's Detection & Brain Segmentation Model | Machine Learning Engineer 2025

- Designed a high-accuracy Convolutional Neural Network (CNN) in Python to classify disease stages from MRI scans, heavily optimizing the model for predictive performance.
- Implemented semantic segmentation to map critical brain structures, deploying the workflow as an automated diagnostic tool.

Publications & Research Projects

- A Unified Deep Learning Framework for Automated Morphological Classification and Photometric Analysis using DECals Legacy Imaging** (MNRAS Positive Review)
 - Galamo is an open-source Python package for comprehensive galaxy analysis, integrating machine learning and statistical methods. It provides automated tools for morphology classification, kinematics, photometry, and spectral analysis to aid astrophysical research.
 - Authors: Jashanpreet Singh Dingra, **Hum Chand (Former Director AIRES, India)** & Madhu Sudan
www.galamo.org

International Collaborations - Computational Astrophysics

- The Evolutionary Destiny and Observational Deficits of JWST “Little Red Dot” Analogues in(On Going - 2026) IllustrisTNG Simulation**
 - By tracking their main descendant branches forward to $z = 0$, we predict these early-universe anomalies are the direct progenitors of today's massive Brightest Cluster Galaxies (BCGs). Furthermore, a stacked temporal analysis aligned to the quenching epoch definitively proves that radiatively efficient AGN feedback is responsible for their rapid star-formation cessation.
 - Authors: Jashanpreet Singh Dingra, **Akanksha Kapathia (Max Planck Institute of Astronomy, Heidelberg, Germany)**
- Exploring Active Galactic Nuclei in the Sloan Digital Sky Survey: A Multiwavelength Study** (2026)

- Project conducted under the **Nicolaus Copernicus Astronomical Center of the Polish Academy of Sciences, Warsaw.**
- *Mentors & Collaborators: Marzena Sniegowska (Astronomical Institute of the Czech Academy of Sciences) and Biswaraj Palit (Copernicus Astronomical Center of the Polish Academy of Sciences)*

Major Computational Astronomy Contributions

- Astropy - Astrophysics and astronomy python package(2025 - Present)
- *Contributed in astropy.coordinates and many more modules.*

Achievements

Academic Grants & Research Infrastructure

- **IUCAA Pegasus Cluster Computational Resource Grant** for JWST LRD Research During my Undergrad. (2026)

Academic Awards

- **ICNPA** (International Conference on Nuclear Physics and Application) **Best Oral Presentation Award.** (2024)
- **JEE Mains** (Joint Entrance Examination) **AIR 231.** (2024)
- **NSEA** (National Standard Examination For Astronomy) **Scholar.** (2023)
- **The Breakthrough Junior Challenge Finalist.** (2021)

Other Awards and Honours

- **ISRO** (Indian Space and Research Organisation) **Space Tutor.** (2023 - Present)

Hackathons & Tech Communities

- **Niohack Runner Up.** - \$500 (2026)
- **AlgOlympia, Chandigarh University** (National Hackathon) **Winner.** - 50000 INR (2026)
- **Shipwrecked Hackathon** (Hack Club, Boston) **Finalist.** (2025)
- **AdventureX** (China’s Biggest Hackathon, Hangzhou) **Top Finalist.** (2025)
- **Ideathon, GNDU** (Golden Jubilee Center of Entrepreneurship and Innovation) **Winner.** - 15000 INR (2025)
- **HACKOWASP 7.0** (North India’s Biggest Hackathon, TIET) **Runner Up.** - 15000 INR (2025)
- **Ad Astra** (National Astronomy Ideathon, NIT, Jalandhar) **Second Runner Up.** - 5000 INR (2025)
- **Hack Club, USA Active Member.** (Present)

Technical Skills

Programming Languages: Python, C++, C, R, FORTAN, MERN, Assembly, MATLAB, Bash
Database Systems: PostgreSQL, MySQL, MongoDB, Redis, GraphQL, Elasticsearch
Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Airflow, Git, Linux
Tech Stack: FastAPI, LangChain, LangGraph, REST API, Weights & Biases, MLflow
Libraries / Frameworks: PyTorch, TensorFlow, JAX, Scikit-learn, HuggingFace, OpenCV, PIL
Machine Learning: Transformers, GNNs, State Space Models (Mamba), RAG, LLM Fine-tuning (SFT, LoRA, QLoRA, PEFT), Agentic AI
Quant Tools: Statsmodels, cvxpy, JAX, TA-Lib, QuantLib, PyPortfolioOpt
Visualization: Matplotlib, Seaborn, Plotly, Power BI, Tableau
Core CS Fundamentals: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA)

External Coursework

NPTEL Computer Science Course	Institution	Year
Discrete Mathematics	IIT Ropar	2024
Data Structures And Algorithms	IIT Delhi	2025
Design & Analysis of Algorithms	IIT Kanpur	2025
Database Management Systems	IIT Madras	2025
Operating Systems	IIT Delhi	2025
Computer Architecture	IIT Delhi	2025